# 5.4 Eigenvectors and linear transformations

Recall our original motivation in studying eigenvectors/eigenvalues. We wanted to understand what a linear transformation $T: \mathbb{R}^n \to \mathbb{R}^n$ is doing to $\mathbb{R}^n$. Since every such linear transformation is given matrix multiplication, i.e.

$$T(v) = Av \quad \text{for all } v \in \mathbb{R}^n \text{ and some } A \in M_{n \times n},$$

we are reduced to studying $A$.

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Q. What about linear transformations $T: V \to V$ where $V \neq \mathbb{R}^n$.

A. When $V$ is a finite-dimensional vector space we can choose a basis $B = \{b_1, \dots, b_n\}$ and use the coordinate map $[\cdot]_B : V \to \mathbb{R}^n$ to convert eigenvector/eigenvalue questions into those we've solved for matrices.

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Def. Let $V$ be a (real) vector space and $T: V \to V$ be a linear transformation. An <u>eigenvector</u> for $T$ is a vector $v \in V$ such that

(i) $v \neq 0$

(ii) $T(v) = \lambda v$

for some $\lambda \in \mathbb{R}$

In this case $\lambda$ is the eigenvalue for $v$.

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Example

Let $V$ be the real vector space of real-valued functions on $\mathbb{R}$, which are infinitely differentiable.

Let $D: V \to V$ be the derivative map.

i.e. $D(f) = f' \qquad \forall f \in V.$

$D$ is a linear transformation.

Let $f \in V$ be the function $f(x) = e^x \ \forall x \in \mathbb{R}$

We know that $D(f) = f' = f$.

$\Rightarrow \quad D(f) = 1 f \quad , \quad f \neq 0.$

i.e. $f$ is an eigenvector of $D$ with eigenvalue 1.

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For the rest of this section $V$ is an $n$-dimensional (real) vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$.

$T: V \to V$ is a linear transformation

Recall, that for any $v \in V$ there exist $c_1, \dots, c_n \in \mathbb{R}$ such that $v = c_1 b_1 + \dots + c_n b_n$ and

$$[v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n.$$

The coordinate map $[\cdot]_B : V \to \mathbb{R}^n$ is an invertible linear transformation, i.e. an isomorphism (Thm 9 §4.4)

Consider the diagram

$$\begin{tikzcd} v \arrow[r, "T"] \arrow[d] & T(v) \arrow[d] \\ {[v]_B} \arrow[r, dashed, "?"'] & {[T(v)]_B} \end{tikzcd}$$

The lower horizontal map is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. It is therefore given by matrix multiplication. What is the matrix?

Again take $v = c_1 b_1 + \dots + c_n b_n$ with $[v]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$.

$$T(v) = T(c_1 b_1 + \dots + c_n b_n)$$

$$= c_1 T(b_1) + c_2 T(b_2) + \dots + c_n T(b_n) \quad (T \text{ is linear})$$

$$\Rightarrow [T(v)]_B = [c_1 T(b_1) + \dots + c_n T(b_n)]_B$$

$$= c_1 [T(b_1)]_B + \dots + c_n [T(b_n)]_B. \quad ([\cdot]_B \text{ is linear})$$

$$= \begin{bmatrix} [T(b_1)]_B & \cdots & [T(b_n)]_B \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad (\text{def'n of matrix mult.})$$

$$= \begin{bmatrix} [T(b_1)]_B & \cdots & [T(b_n)]_B \end{bmatrix} [v]_B.$$

$$\Rightarrow [T(v)]_B = \left[ [T(b_1)]_B \cdots [T(b_n)]_B \right] [v]_B.$$

The question mark in the above diagram is matrix multiplication by $\left[ [T(b_1)]_B \cdots [T(b_n)]_B \right] \in M_{n \times n}$

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Def. The <u>matrix for the linear transformation</u> $T: V \to V$ (relative to $B$) is

$$[T]_B = \left[ [T(b_1)]_B \cdots [T(b_n)]_B \right]$$

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With this notation $[T(v)]_B = [T]_B [v]_B$, $\forall v \in V$

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Example

Let $V = \mathbb{P}_2$ with basis $B = \{1, x, x^2\}$.

and let $T(p) = p'$ $\forall$ $p \in \mathbb{P}_2$.

Then $T: \mathbb{P}_2 \to \mathbb{P}_2$ is a linear transformation.

$$[T]_B = \left[ [T(1)]_B \ [T(x)]_B \ [T(x^2)]_B \right]$$

Let's compute...

$$T(1) = 0 = 0(1) + 0(x) + 0(x^2)$$

$$\Rightarrow [T(1)]_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$T(x) = 1 = 1(1) + 0x + 0x^2$$

$$\Rightarrow [T(x)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$T(x^2) = 2x = 0(1) + 2x + 0x^2$$

$$\Rightarrow [T(x^2)]_B = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

So $[T]_B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

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Remarks.

Recall the general theorem about isomorphic vector spaces. It says that any fact about linear algebra in one vector space is true in the other isomorphic vector space. We use the isomorphism to convert a fact from one vector space into a fact about the other one.

In the case at hand $V$ is isomorphic to $\mathbb{R}^n$ with isomorphism $[\cdot]_B : V \to \mathbb{R}^n$. How does this general theorem apply to eigenvectors/eigenvalues?

Suppose $v \in V$ is an eigenvector for $T$ i.e.

i.e. $v \neq 0$ and $T(v) = \lambda v$.

Then $[T(v)]_B = [\lambda v]_B$

$$\Leftrightarrow [T(v)]_B = \lambda [v]_B.$$

$\iff [T]_B [v]_B = \lambda [v]_B$

$\iff$ $[v]_B \in \mathbb{R}^n$ is an eigenvector for the matrix $[T]_B$ with eigenvalue $\lambda$.

We can reverse this process by using the inverse map $[\cdot]_B^{-1} : \mathbb{R}^n \to V$ of $[\cdot]_B$.

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Theorem

The eigvalues of $T$ are the eigenvalues of $[T]_B \in M_{n \times n}$. Every eigenvector $v$ of $T$ corresponds to an eigenvector $[v]_B$ of $[T]_B$.

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Remark. This theorem works for <u>any</u> basis $B$ of $V$.

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Def (Not in textbook).

The <u>characteristic polynomial</u> of $T$ is the characteristic polynomial of $[T]_B$.

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Example

Take $V = \mathbb{P}_2$ and $T(p) = p'$ as in previous example.

Eigenvalues of $T$ = eigenvalues of $[T]_B$

= roots of $\det([T]_B - xI)$

$$\det\left(\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} - xI\right) = \det\begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 2 \\ 0 & 0 & -x \end{pmatrix}$$

$$= (-x)^3.$$

Clearly $-x^3 = 0 \iff x = 0$ so $T$ only has one eigenvalue $\lambda = 0$.

The $0$-eigenspace of $[T]_B$ is

$$\begin{aligned} \mathrm{Nul}([T]_B - 0I) &= \mathrm{Nul}([T]_B) \\ &= \mathrm{Nul}\left(\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}\right) \\ &= \mathrm{Nul}\left(\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\right) \\ &= \left\{ \begin{bmatrix} c_1 \\ 0 \\ 0 \end{bmatrix} : c_1 \in \mathbb{R} \right\} \\ &= \mathrm{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\}. \end{aligned}$$

$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is an eigenvector of $[T]_B$ with eigenvalue $0$.

The vector $p \in \mathbb{P}_2$ with $[p]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is

$$p(x) = 1(1) + 0x + 0x^2 = 1$$

We have proved that $T(p) = 0$
and the only polynomials with 0 eigenvalue
are scalar multiples of $p(x)=1$. i.e. constant functions.